

overview

the module, the project, and where everything belongs · prof. dr. nicolas meseth

- 1 why this module
- 2 what you will learn
- 3 the project
- 4 experiment: you are the receiver
- 5 1 + 4 challenges
- 6 how we work
- 7 this semester

📊 quick poll

have you ever written a program?

yes, quite a lot

a little, at school or for fun

never



lfi.uber.space/q

📊 quick poll

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yes, quite a lot



a little, at school or for fun



never



lifi.uber.space/q

📊 quick poll

how often do you use ai chats like ChatGPT?

every day

every week

now and then

never



lfi.uber.space/q

📊 quick poll

how often do you use ai chats like ChatGPT?

every day



every week



now and then



never



lfi.uber.space/q

📊 quick poll

have you ever worked with an AI agent, like Claude Code, Codex or OpenCode?

yes

i have heard of them

no, never heard of them



lfi.uber.space/q

📊 quick poll

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lfi.uber.space/q

📊 quick poll

how well do you know what happens inside a computer?

very well

roughly

hardly

not at all



lfi.uber.space/q

📊 quick poll

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hardly



not at all



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how many different values can one byte hold?

8

16

256

no idea



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how many different values can one byte hold?

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☐

16

☐

256

☒

no idea

☐

lifi.uber.space/q

how many different values can one byte hold?

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☐

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256

☒

no idea

☐

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📱 quick quiz

a photo on your phone: roughly how many numbers are stored for it?

a few hundred

a few thousand

tens of millions

no idea



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a photo on your phone: roughly how many numbers are stored for it?

a few hundred

☐

a few thousand

☐

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☐

no idea

☐

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a photo on your phone: roughly how many numbers are stored for it?

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a few thousand



tens of millions



no idea



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📄 quick quiz

at the lowest level, what does a computer work with?

words

pictures and sounds

numbers: zeros and ones

no idea



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at the lowest level, what does a computer work with?

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at the lowest level, what does a computer work with?

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📄 quick quiz

you switch off your computer. what happens to what is in its ram?

it stays there

it is lost

no idea



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you switch off your computer. what happens to what is in its ram?

it stays there

☐

it is lost

☐

no idea

☐

lifi.uber.space/q

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no idea

☐
☒
☐

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📊 quick poll

how important will it be for your career to understand AI
and to work well with it?

essential

very important

somewhat important

not important



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📊 quick poll

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and to work well with it?

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part 1

why this module

why this module

AI answers any question in seconds.

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it writes, summarises, translates and explains. it seems to know everything.

why this module

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or does it?

AI answers any question in seconds.
it writes, summarises, translates and explains. it seems to know everything.
or does it?

so why learn anything at all?

the same question, fifty years ago:

everyone has
a calculator.
why still
learn maths?



image: AI-generated (gpt-image-2.5-flare)

why this module

why maths, if you have a calculator?

to write the problem down so that the calculator can solve it.

why maths, if you have a calculator?

to write the problem down so that the calculator can solve it.
to notice when 4,000 comes out and 40 would be right.

why maths, if you have a calculator?

to write the problem down so that the calculator can solve it.
to notice when 4,000 comes out and 40 would be right.
to know what it cannot calculate at all.

why maths, if you have a calculator?

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to notice when 4,000 comes out and 40 would be right.

to know what it cannot calculate at all.

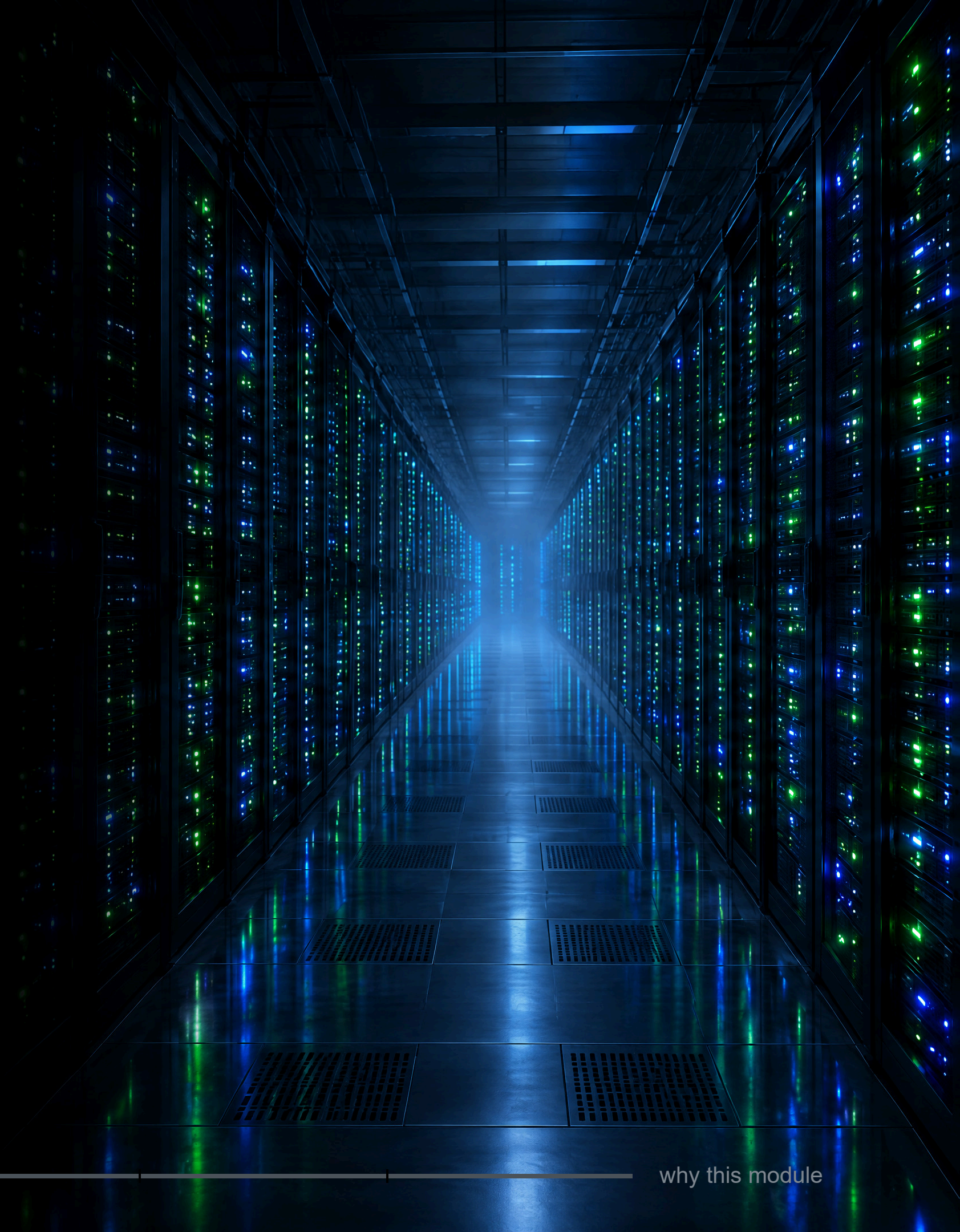
the same holds for computers, and for AI.

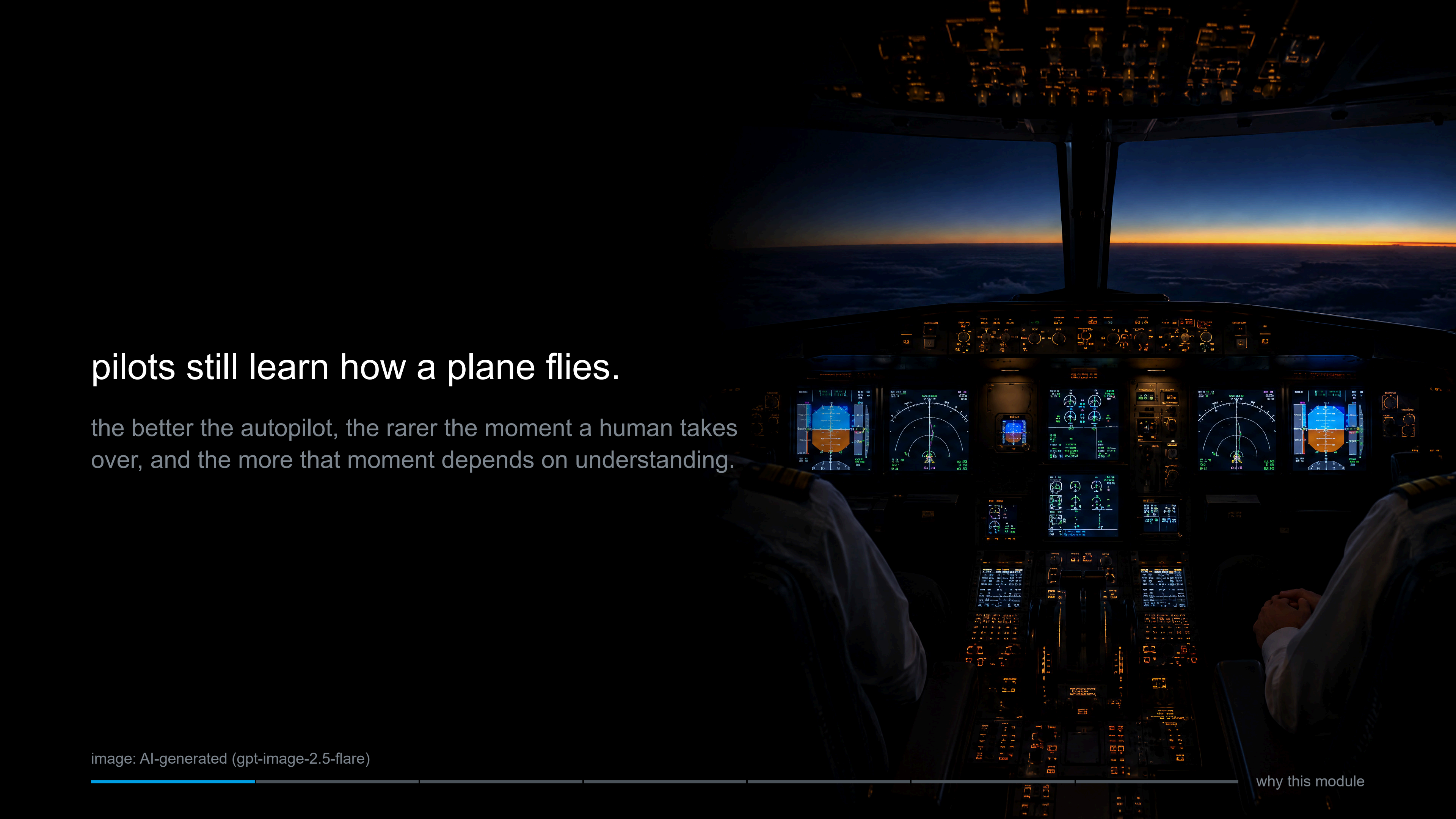
a computer is just a bigger calculator.

it turns numbers into numbers, billions of times a second. to
judge what comes out, you have to understand what it does.

image: AI-generated (gpt-image-2.5-flare)

why this module





pilots still learn how a plane flies.

the better the autopilot, the rarer the moment a human takes over, and the more that moment depends on understanding.

the most powerful calculator humans built so far

it answers anything in seconds.



image: AI-generated (gpt-image-2.5-flare, edited with gpt-image-2.5-sunburst)

why this module

the most powerful calculator humans built so far

it answers anything in seconds.
it sounds right, even when it is wrong.



image: AI-generated (gpt-image-2.5-flare, edited with gpt-image-2.5-sunburst)

why this module

the most powerful calculator humans built so far

it answers anything in seconds.
it sounds right, even when it is wrong.
it cannot check its answer against the
real world.



image: AI-generated (gpt-image-2.5-flare, edited with gpt-image-2.5-sunburst)

why this module

the most powerful calculator humans built so far

it answers anything in seconds.
it sounds right, even when it is wrong.
it cannot check its answer against the
real world.

you learn to work with it: cut, check,
measure.



image: AI-generated (gpt-image-2.5-flare, edited with gpt-image-2.5-sunburst)

and to understand AI

AI

is software: a program, like any other.

and to understand AI

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is software: a program, like any other.

software

runs on a computer.

and to understand AI

AI

is software: a program, like any other.

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turns numbers into numbers, billions of times a second.

and to understand AI

AI

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software

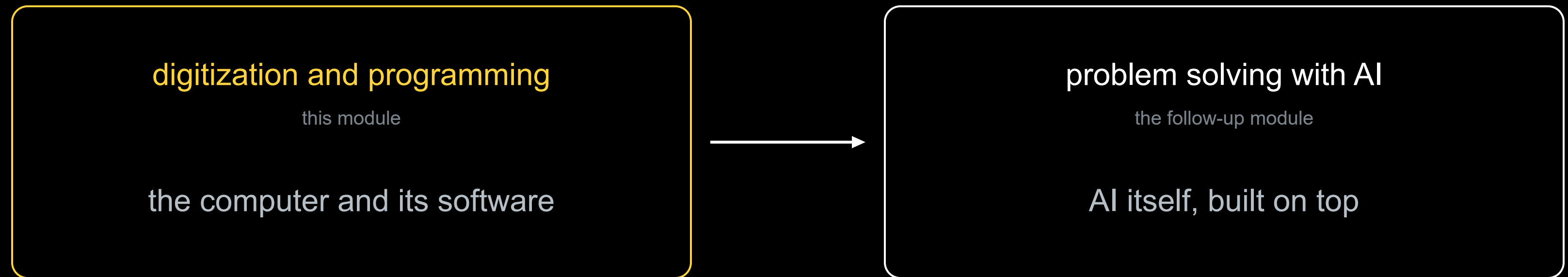
runs on a computer.

computer

turns numbers into numbers, billions of times a second.

to understand AI, you first understand the computer and its software.

what comes next



this module is the foundation. the next one builds on it.

📊 quick poll

would you like to have a better understanding of computers?

yes, definitely

a little

not really



lifi.uber.space/q

would you like to have a better understanding of computers?

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part 2

what you will learn

what you will learn

how do computers solve complex problems?

how do computers solve complex problems?

how do computers
represent information?

how do computers
store information?

how do computers
transfer information?

how do computers
process information?

how do computers solve complex problems?

how do computers
represent information?

making bits

how do computers
store information?

storing bits

how do computers
transfer information?

transferring bits

how do computers
process information?

processing bits

how do computers solve complex problems?

how do computers
represent information?

making bits

analog and digital

symbols and information

number systems

code systems

how do computers
store information?

storing bits

memory and storage

how do computers
transfer information?

transferring bits

signal and noise

sampling and sync

protocols

errors and redundancy

throughput and limits

how do computers
process information?

processing bits

logic and arithmetic

compression

encryption

- cutting problems
- input, processing, output
- algorithms and programs
- measuring and experimenting
- abstraction and layers

how do computers solve complex problems?

how do computers represent information?

- making bits
- analog and digital
- symbols and information
- number systems
- code systems

how do computers store information?

- storing bits
- memory and storage

how do computers transfer information?

- transferring bits
- signal and noise
- sampling and sync
- protocols
- errors and redundancy
- throughput and limits

how do computers process information?

- processing bits
- logic and arithmetic
- compression
- encryption

the map lives on the website

LiFi Project

Concepts > Overview

Overview

The module, the project, and where everything belongs

This page is the map. It shows how the module is put together: the one question behind it, the four questions underneath, the eighteen concepts and their five colours, and the five challenges of the project with the concepts each of them needs. You will see this map again and again during the semester, on the slides and on this page, each time with the part lit up that we are working on. If you ever lose track of where you are, come back here.

One question, four questions

The module has one big question: **how can we solve complex problems with computers?** To answer it you need to understand what a computer does with information, and there are four things it does. It represents information, it stores it, it transfers it, and it processes it.

cutting problems

input, processing, output

algorithms and programs

measuring and experimenting

abstraction and layers

how can we solve complex problems with computers?

how do computers represent information?

making bits

analog and digital

symbols and information

number systems

code systems

how do computers store information?

storing bits

memory and storage

how do computers transfer information?

transferring bits

signal and noise

sampling and sync

protocols

errors and redundancy

throughput and limits

how do computers process information?

processing bits

logic and arithmetic

compression

encryption

what you will learn

part 3

the project

why a project,
and not just
lectures?



image: AI-generated from photos of the real device (gpt-image-2.5-sunburst)

the project

why a project?



learn by doing

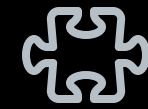
concepts stick when you have
built them, broken them and
fixed them yourself.

why a project?



learn by doing

concepts stick when you have
built them, broken them and
fixed them yourself.



problems first

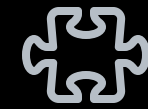
every problem meets you on
your own device before it gets
a name in class.

why a project?



learn by doing

concepts stick when you have built them, broken them and fixed them yourself.



problems first

every problem meets you on your own device before it gets a name in class.



reality answers

a device does not care how good an answer sounds. it works, or it does not.

the goal of the lifi project

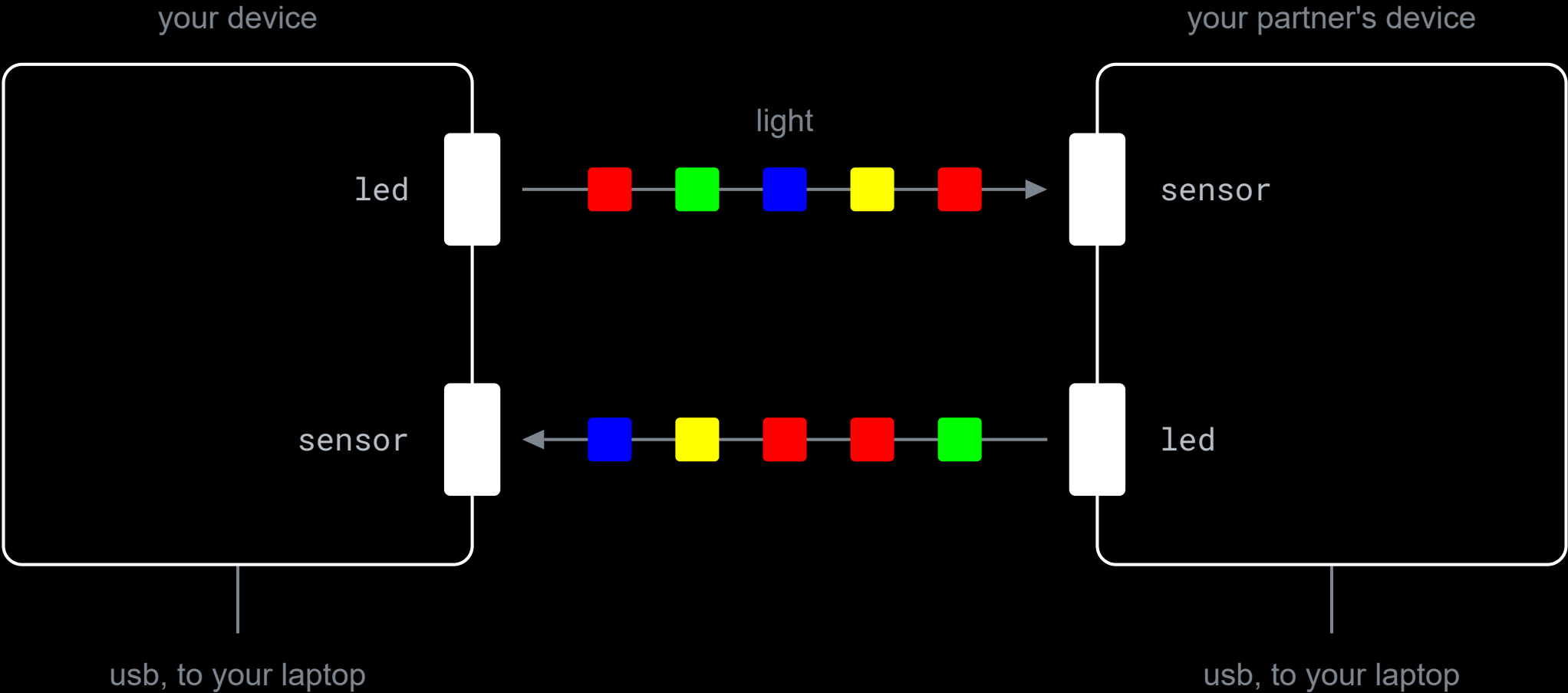
there is a photo
on this laptop.
how does it get
to the other one?



image: AI-generated (gpt-image-2.5-flare)

the project

the device



a lamp and an eye on the front of each device. two devices face each other, so light travels both ways.

why a lamp

everything interesting in a computer is
invisible.

this project makes it visible: a few times a second, on your
desk, where you can explore it with your own hands.

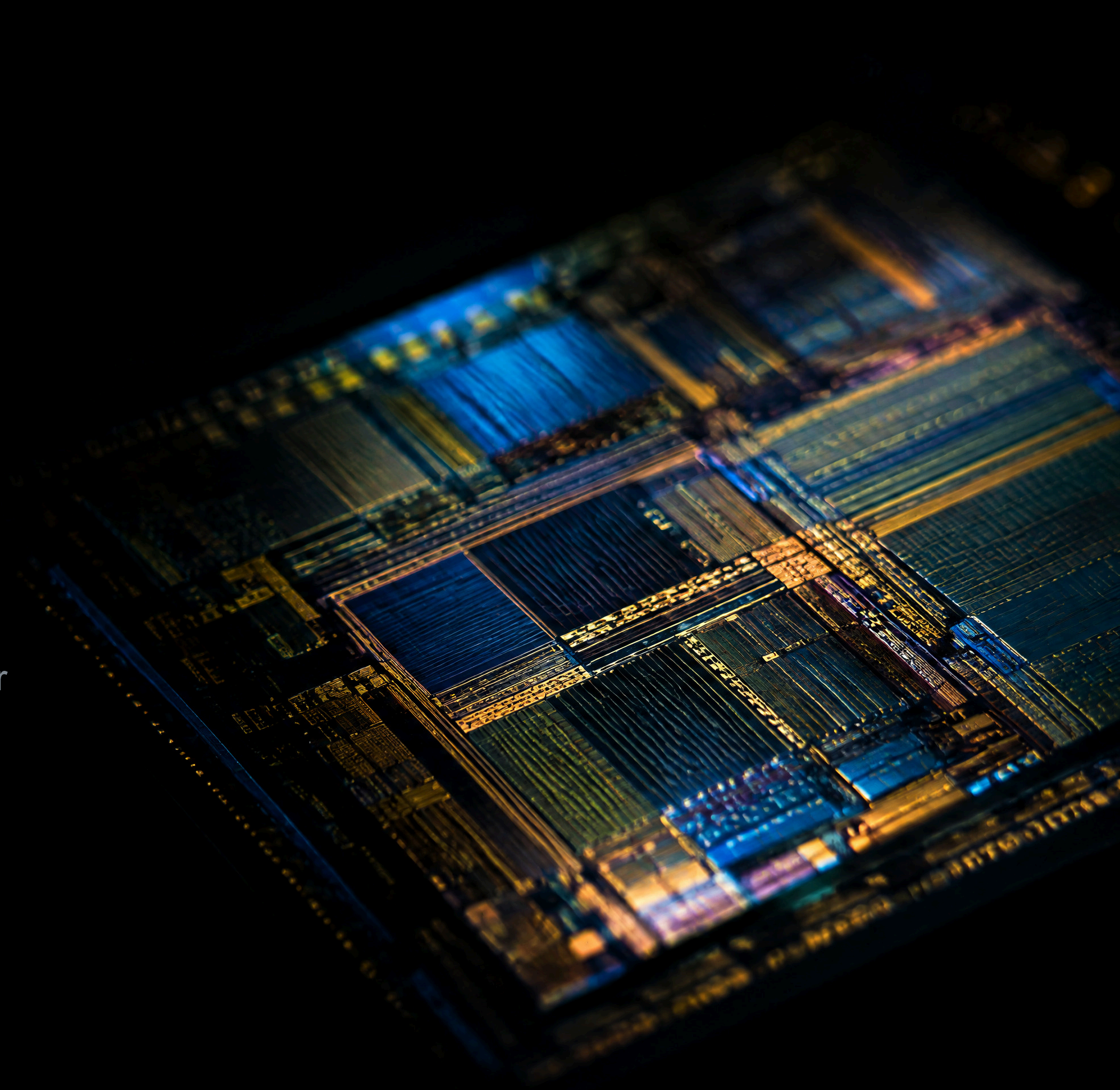


image: AI-generated (gpt-image-2.5-flare)

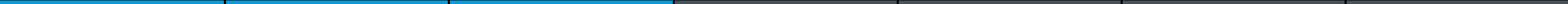
the project

the journey of one photo



a photo on your laptop

memory and storage



the journey of one photo



a photo on your laptop



becomes bytes

memory and storage

number systems

code systems

the journey of one photo



a photo on your laptop



becomes bytes



is packed into a frame with a checksum

memory and storage

number systems

code systems

protocols

errors and redundancy

compression

encryption

the journey of one photo



a photo on your laptop



becomes bytes



is packed into a frame with a checksum



bytes become colours

memory and storage

number systems

code systems

protocols

errors and redundancy

compression

encryption

symbols and information

analog and digital

the journey of one photo



a photo on your laptop



becomes bytes



is packed into a frame with a checksum



bytes become colours



travels as light

memory and storage

number systems

code systems

protocols

errors and redundancy

compression

encryption

symbols and information

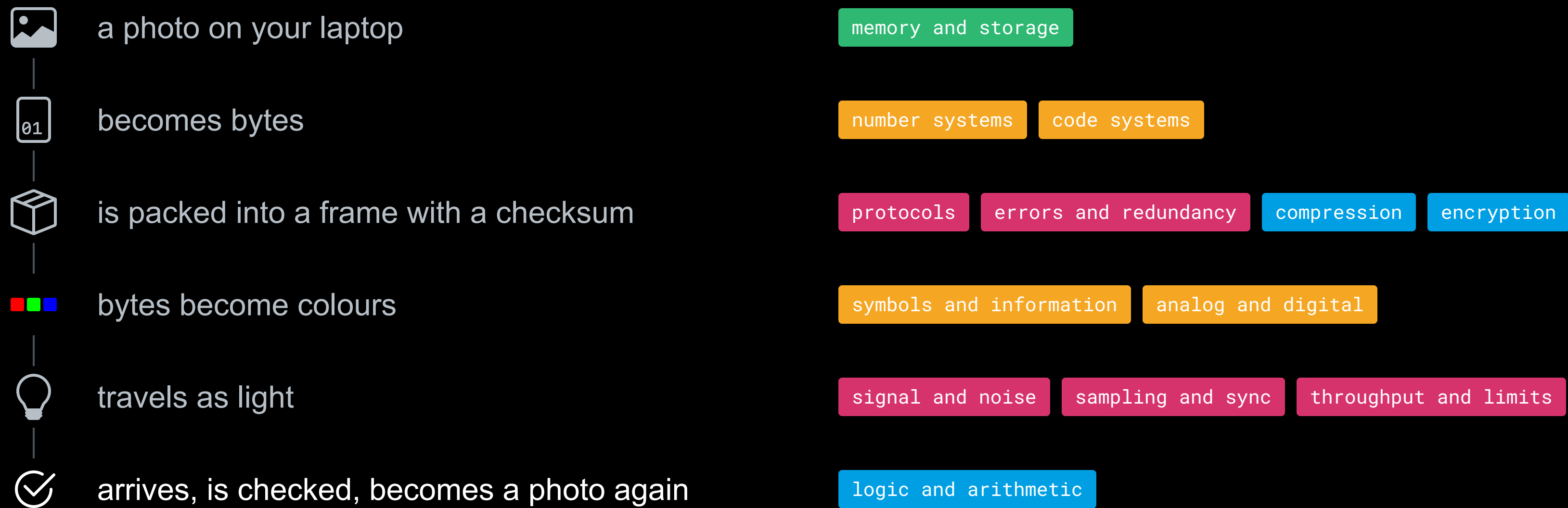
analog and digital

signal and noise

sampling and sync

throughput and limits

the journey of one photo



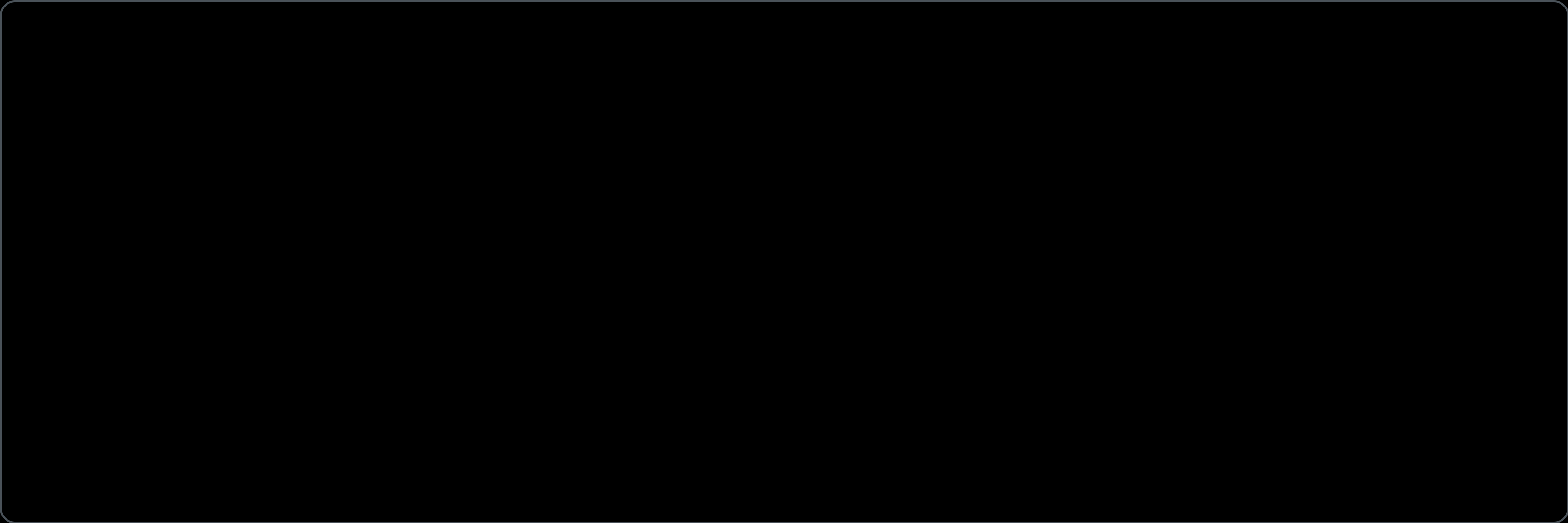
every stop on the way is a concept of this module.

part 4

experiment: you are the receiver

take a pen and a piece of paper.
the screen is the lamp. you are the sensor.
write down every symbol you see, using the alphabet at the bottom.

round 1: four colours, slowly



alphabet



r



g

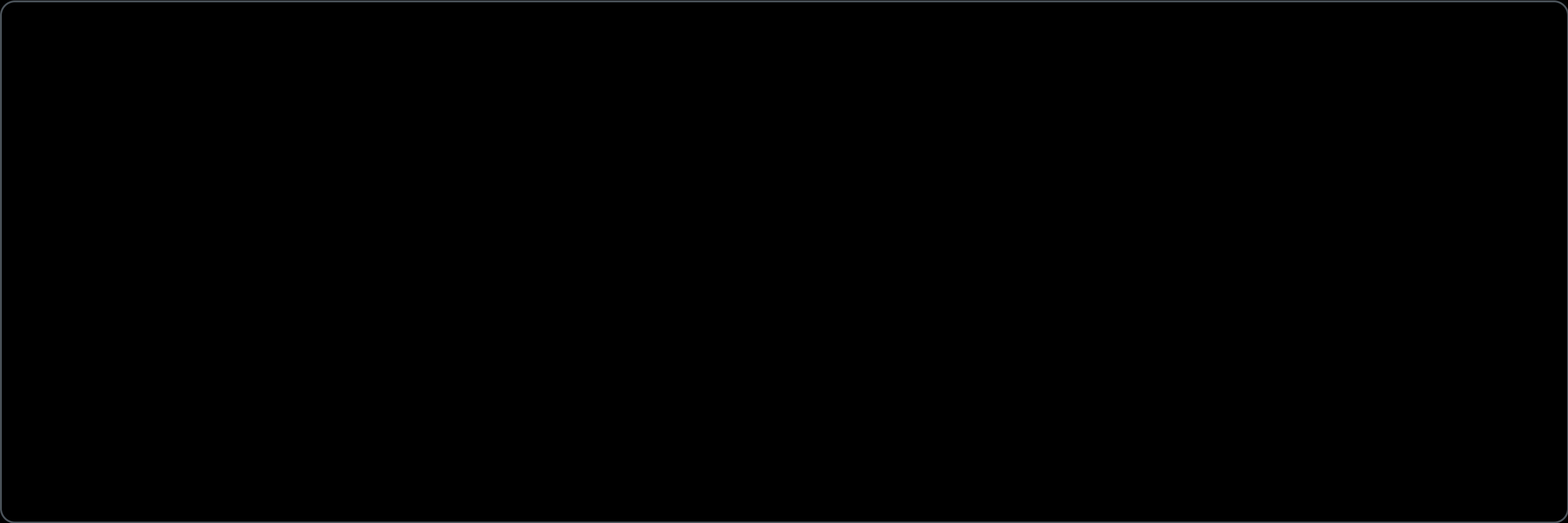


b



y

round 1: four colours, slowly



sent g r y b r y g b

alphabet  r  g  b  y

round 2: eight blues

alphabet

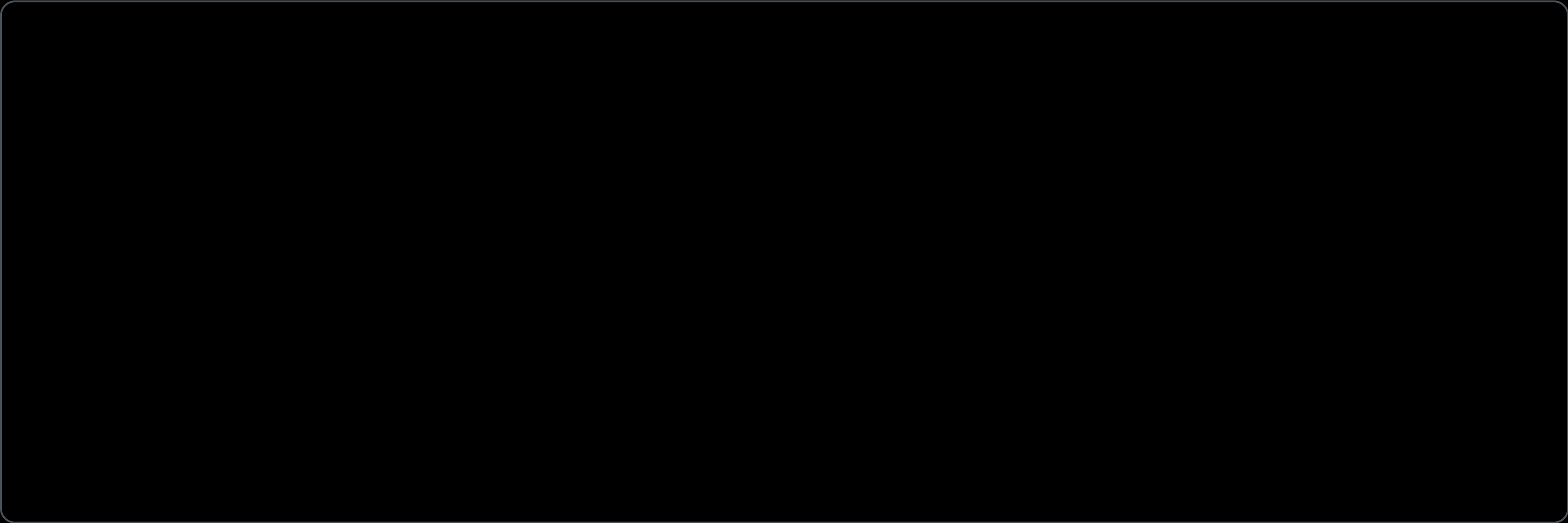
A horizontal bar chart with 8 bars. The bars are arranged horizontally and increase in height from left to right. The colors transition from dark blue on the left to light blue on the right. Each bar is labeled with a number from 1 to 8.

Bar Number	Approximate Height (relative)	Color
1	1	Dark Blue
2	2	Dark Blue
3	3	Dark Blue
4	4	Dark Blue
5	5	Dark Blue
6	6	Dark Blue
7	7	Medium Blue
8	8	Light Blue

experiment: you are the receiver

3 7 1 5 8 2 6 4

round 3: faster



alphabet



r



g

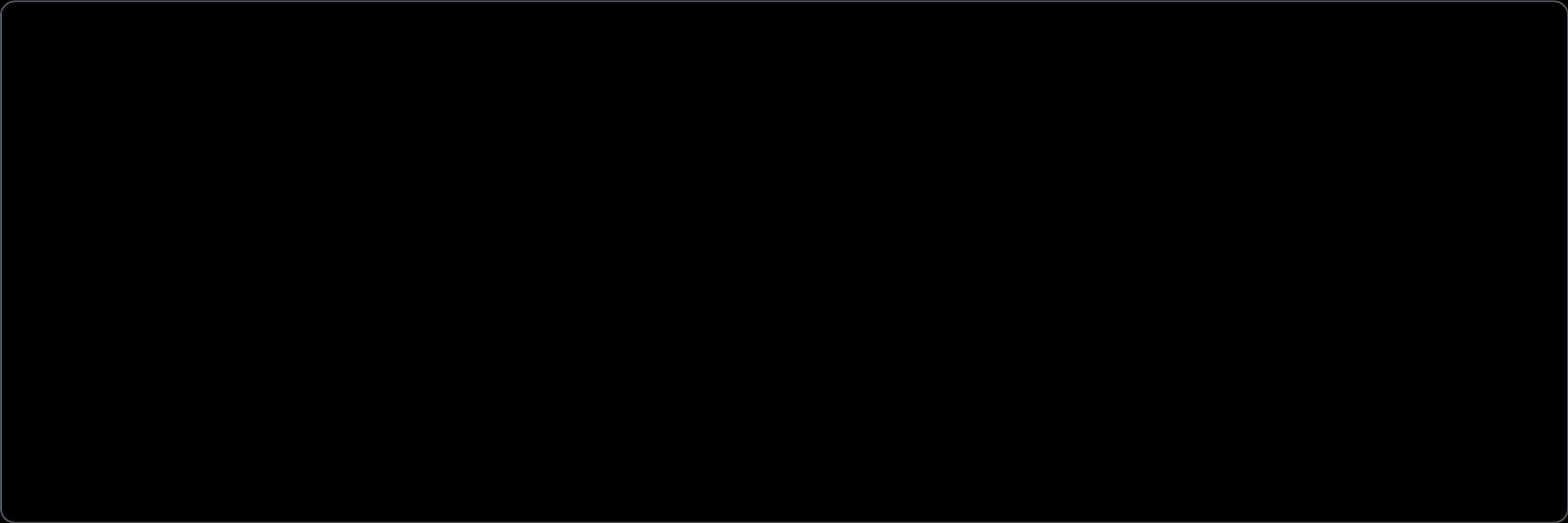


b



y

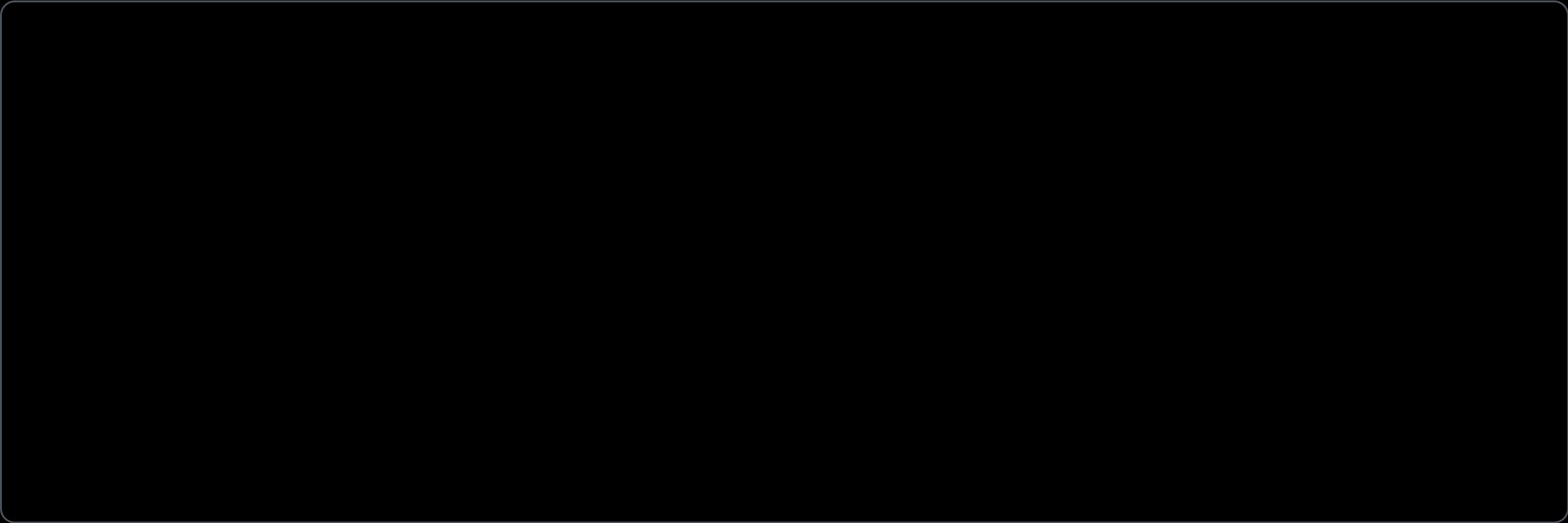
round 3: faster



sent r y g g b r y b g r b y

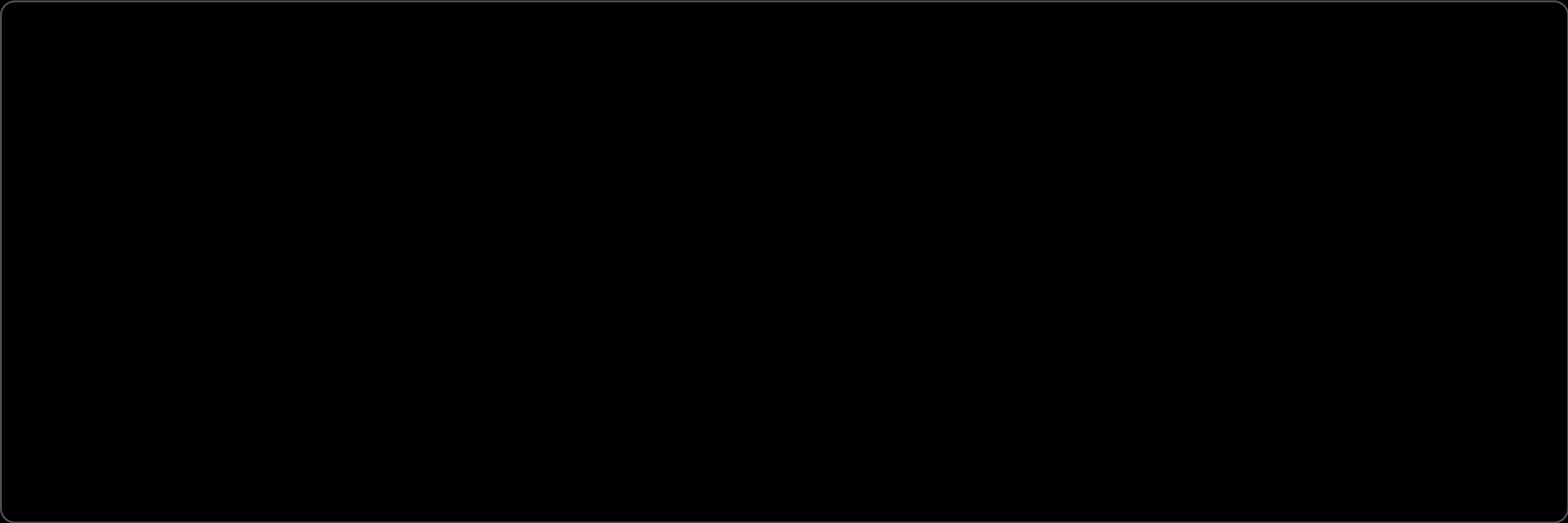
alphabet ■ r ■ g ■ b ■ y

round 4: where is the message?



alphabet  r  g  b  y

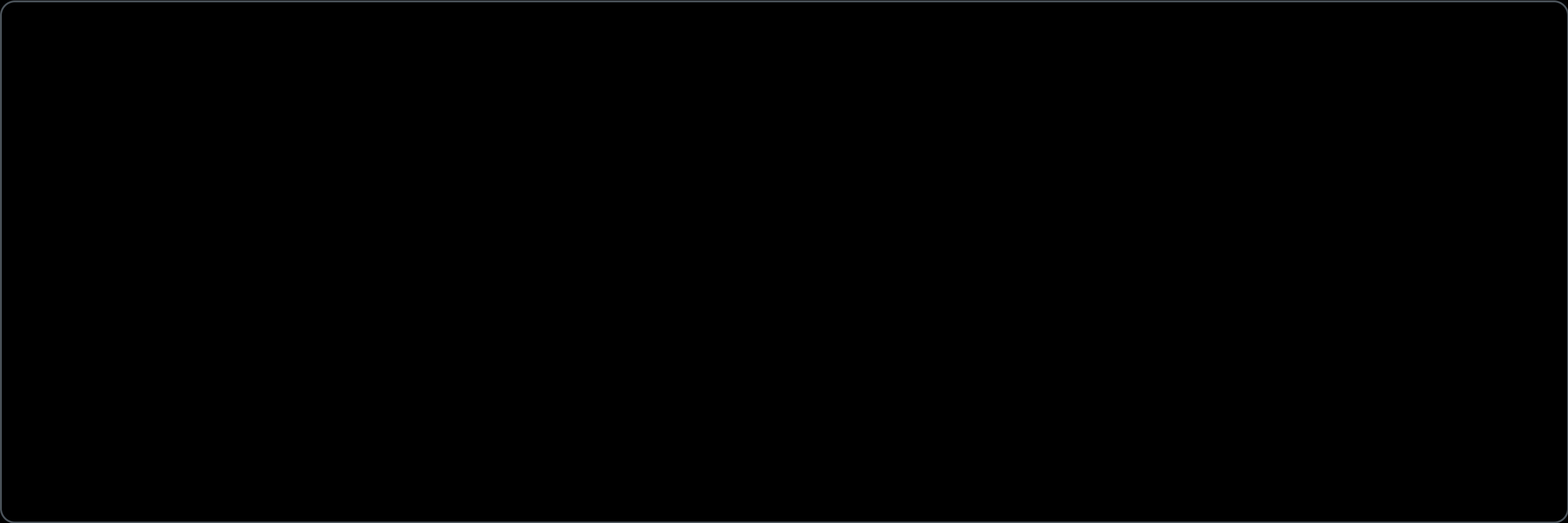
round 4: where is the message?



sent g b r y g r

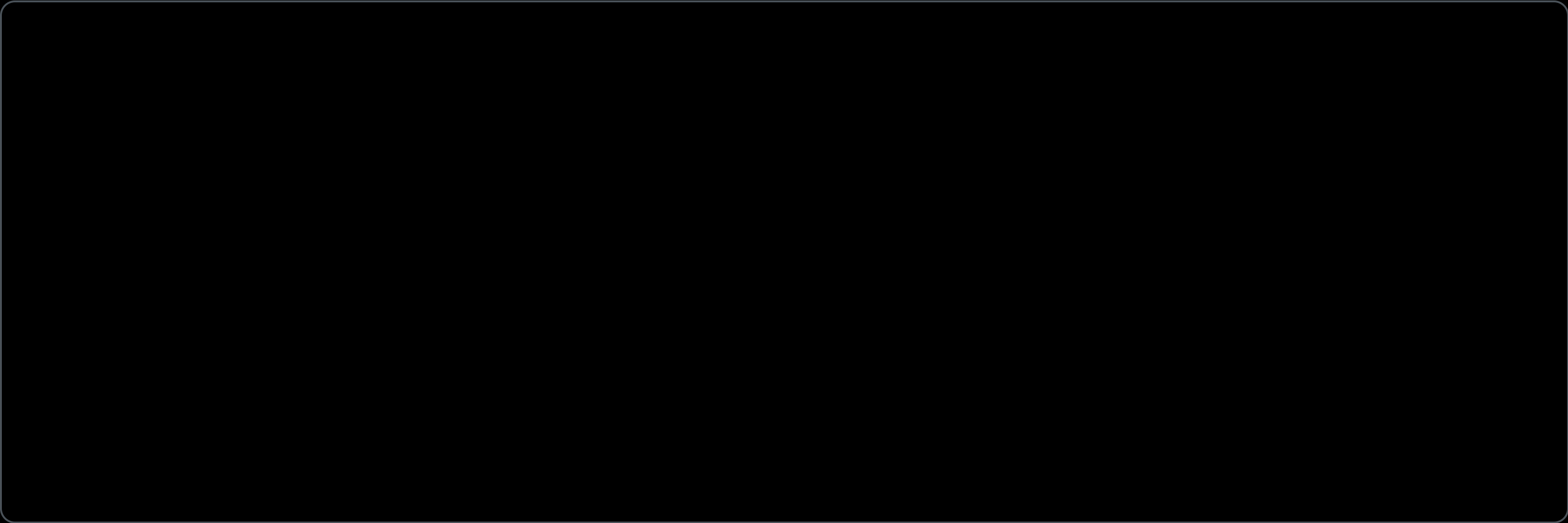
alphabet  r  g  b  y

round 5: did everything arrive?



alphabet ■ r ■ g ■ b ■ y

round 5: did everything arrive?



sent y g r b g y b r

alphabet  r  g  b  y

part 5

1 + 4 challenges

one new difficulty at a time

- | | | |
|---|--------------|--|
| 0 | the spark | can you bring your device to life? |
| 1 | the alphabet | how many signals can your receiver tell apart? |
| 2 | the word | can you send a whole word, and how fast? |
| 3 | the listener | can the receiver find the start on its own? |
| 4 | the packet | can you send a real file, correct and fast? |

each challenge adds exactly one new difficulty. everything else stays as you know it.

what every challenge produces



measurement logs

what did you change, what did you keep the same, and what came out?



a description of your solution

enough for another team to rebuild it. from challenge 3 on, a full protocol specification.



a log of mistakes

where the assistant suggested something, and a measurement proved it wrong.

nothing of this is handed in or graded. it is for your team.

part 6

how we work



two students, two devices

one device each, so you own
both ends of the link. split the
work, and swap now and
then.



two students, two devices

one device each, so you own
both ends of the link. split the
work, and swap now and
then.



the lamp has the last word

your assistant knows nothing
about your device. measure
before, measure after, and
note where it was wrong.



two students, two devices

one device each, so you own both ends of the link. split the work, and swap now and then.



the lamp has the last word

your assistant knows nothing about your device. measure before, measure after, and note where it was wrong.



try the demonstrators

a small toy in the browser for almost every concept: switch a byte, break a file, add noise.

part 7

this semester

fourteen mondays

1 **kick-off: overview & setup**

devices, setup, challenge 0 starts

2 cutting problems, algorithms and programs

challenge 0: your first program

3 measuring and experimenting

challenge 0 check

4 analog and digital, symbols and information

challenge 1 starts

5 signal and noise

🏆 competition: challenge 1

6 number systems and code systems

challenge 2 starts

7 sampling and synchronization

challenge 2: faster

8 throughput and limits

🏆 competition: challenge 2

9 protocols

challenge 3 starts

10 logic and arithmetic

interop test

11 memory and storage

🏆 competition: challenge 3

12 errors and redundancy

standardisation session, challenge 4 starts

13 compression

challenge 4: faster and shorter

14 abstraction and layers

🏆 final: challenge 4

a typical monday, 15:00 to 18:15

15:00 look back: where are you stuck?

15:30 input: one concept

16:00 experiment, all together

16:30 break

16:45 **workshop**

18:00 outlook and your task for the week

every monday, the longest block is yours: you build.

every monday



room HD0001

mondays, 15:00 to 18:15.



laptop and device

bring both, every time. charge
the laptop before you come.



attendance list

sign it every week. it is for
statistics, not for grades.

the exam



60 minutes

multiple choice on the
concepts, everyone on their
own. the project is where you
meet them.



60 minutes

multiple choice on the
concepts, everyone on their
own. the project is where you
meet them.



+5 % bonus

if all five of your challenges
are accepted. all or nothing.





60 minutes

multiple choice on the concepts, everyone on their own. the project is where you meet them.



+5 % bonus

if all five of your challenges are accepted. all or nothing.



practise

with the questions on every concept page. same format as the exam.

where to find what

	docs.lifi-project.de	everything to read: concepts, challenges, setup
	lms.hs-osnabrueck.de	announcements, changes, the schedule
	your course folder	your assistant and your code
	lifi.uber.space/cockpit	your link, live
	winf-hsos.github.io/lifi-concept-demos	the demonstrators

📱 quick poll

how do you feel about this module?

looking forward to it

curious

a bit worried

worried



lfi.uber.space/q

how do you feel about this module?

looking forward to it

curious

a bit worried

worried



lifi.uber.space/q

by the end of term,
a picture crosses that gap.
and along the way,
more than one light will go on.

